TURKU ENERGIA

# Tasehinta.

Know when the imbalance price breaks away from spot, and why, before it happens.




Team EDISS #1 · Since AI 2026 · Turku Energia challenge

# Every missed plan is settled at a price nobody knows yet.

Finland's imbalance price is set every 15 minutes. It follows spot until the grid runs short of power or into a surplus, then it can jump far either way.

## 4,000 €/MWh

the imbalance price on 9 Sep at 20:30, with spot at 200

## 1 in 4

quarter-hours in September ended more than 50 €/MWh from spot

## 15 min

settlement period: every quarter-hour is its own price

Data: organisers' validation month, 7 Sep – 4 Oct 2026. Photo: Kashima Power Line 07 by Σ64, CC BY 4.0, Wikimedia Commons.



# Every hour: the next 40 quarter-hours, four answers each.

TURKU ENERGIA

### What will it cost?

A median imbalance price and an honest 80 % range around it

### Which way is the grid going?

The probability of up-regulation, down-regulation or none

### Will it spike?

The chance of a price more than 50 €/MWh above or below spot

### Why?

The few factors that move the forecast most, strongest first

**Then it shows only what deserves attention** and says what to check, never what to trade: open data can't see Turku Energia's own position.

# One model, built only from what was known at the time.

TURKU ENERGIA



## Published data

Only what was out before the hour: market, balancing and grid data, Fingrid real-time, weather runs, 19 months of history

## Features

For each of the 40 quarter-hours: recent state, real-time grid, wind-forecast revisions, the shape of the day-ahead price

## One ensemble model

Two gradient-boosting components, averaged: their errors partly cancel

## Forecast + dashboard

The submitted forecast file and the situational view

**Forecast the gap, not the price.** Spot is known the day before; the unknown is how far the imbalance price lands from it.

**Direction first.** Short of power: above spot. Surplus: below. Price, range, direction and spike risk tell one story.

**History made comparable.** Pre-June prices rebuilt under the June 2026 rule: 99.5 % exact where it can be checked.

**Nothing from the future.** Every input used only from its publication time, enforced by automated leak tests.

# More accurate than the baseline on every score.

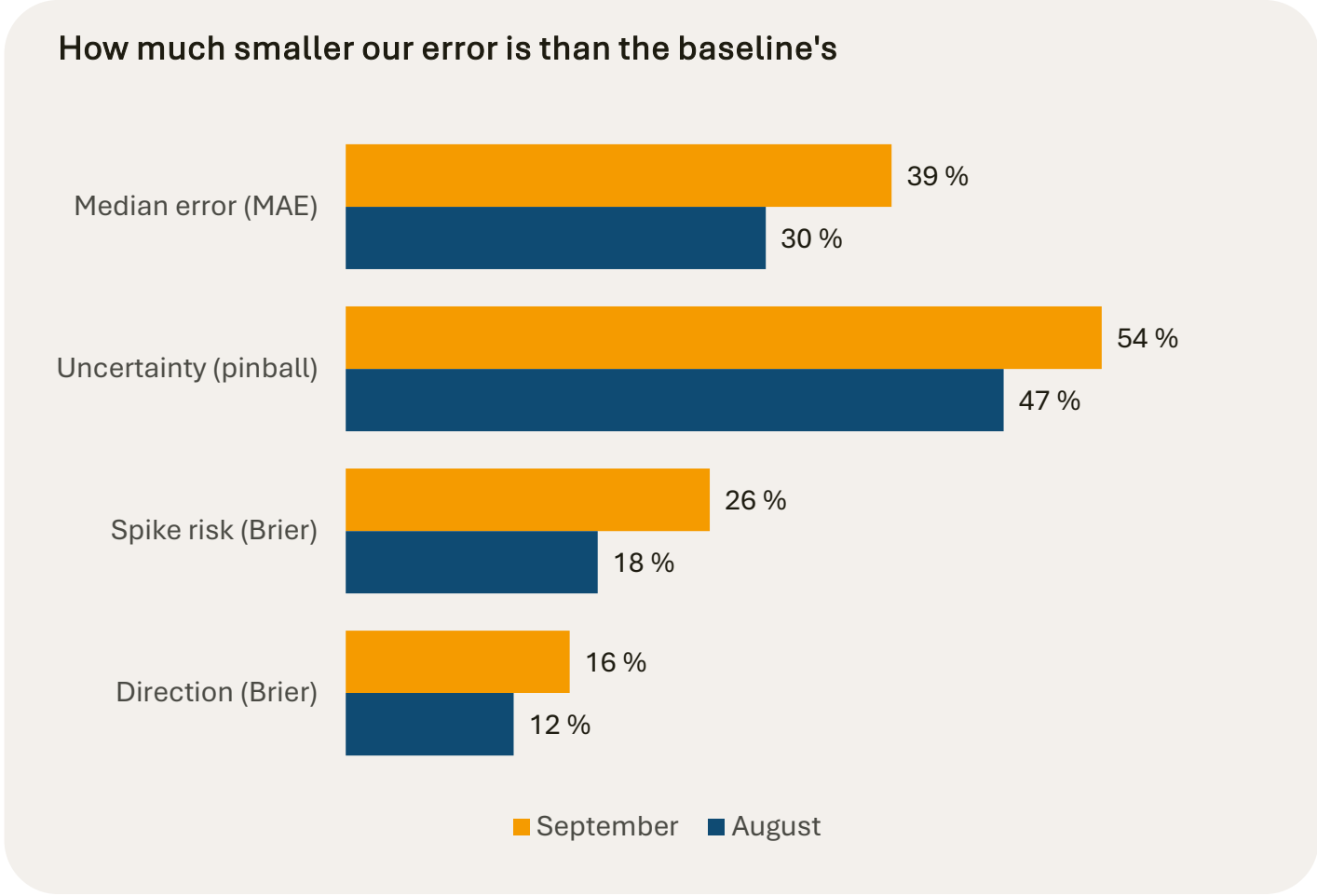
**36.7 %**

better than the organisers' baseline on their validation month (Sep), weighted by the challenge's points

**29.0 %**

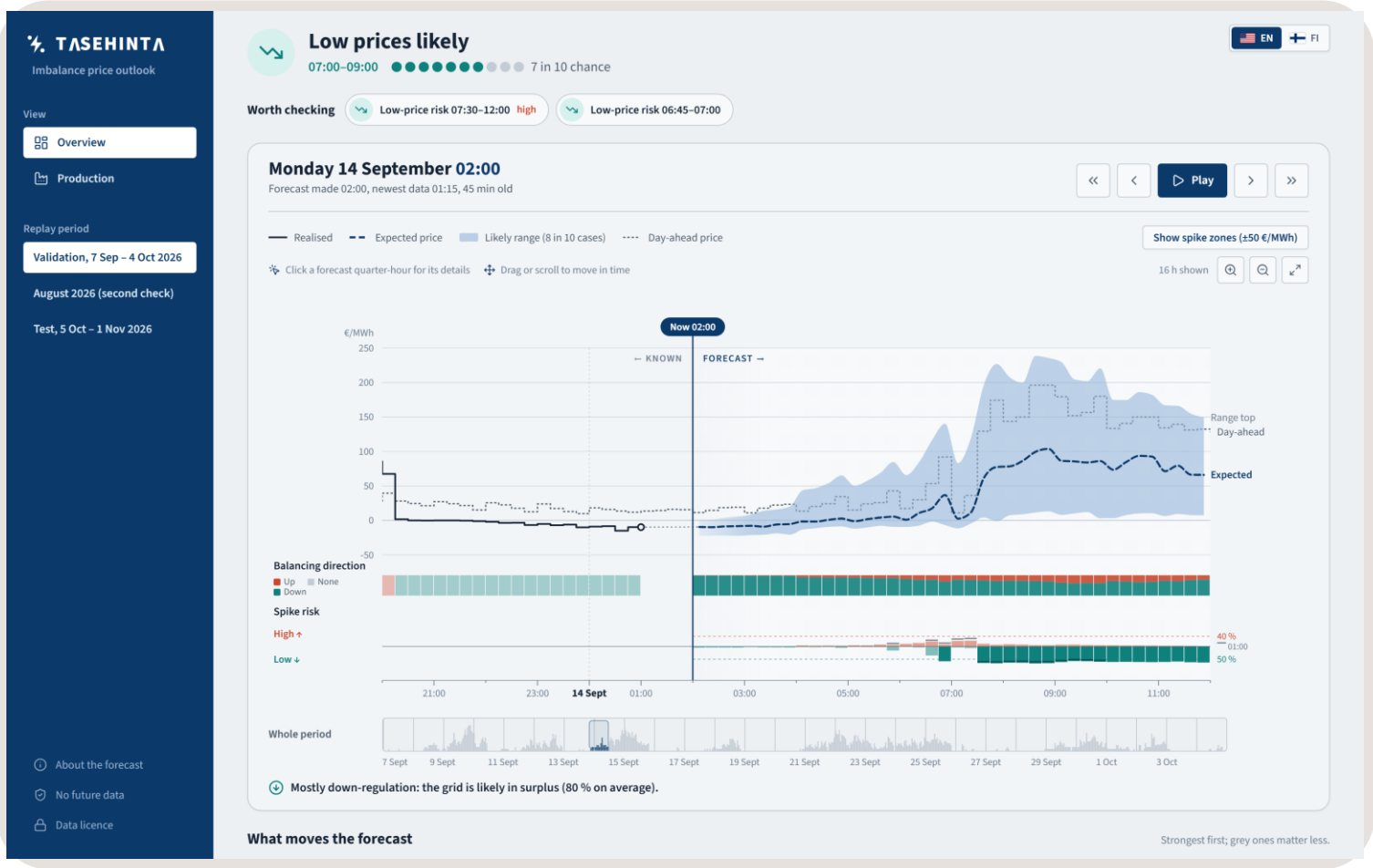
on August, a second month the model never saw

Median error 40.2 vs 66.2 €/MWh · better in every September week · the 80 % range held 81 % of prices




Scored with the organisers' own score.py; each month forecast using only data published before it.

# See what's worth checking, and replay any past hour.




## Only what matters

Spike risk up, regulation expected, forecast moved, an uncertain signal

## One time axis

Price and range, direction and spike risk on one 15-minute grid

## Move through time

Scroll back in 15-minute steps; nothing published later is shown

Live, in English and Finnish:
tasehinta.lovable.app

# From “check this” to “offer this”.



## Turku Energia's own data

Plant capacity, marginal costs and the market position turn a warning into a concrete capacity suggestion.

## Winter weather history

The history covers winter 2025–26, but weather-forecast runs only from April 2026; adding winter runs is the next model step.

## Live operation

Run the same pipeline every hour on live Fingrid and weather data, behind an API.

TURKU ENERGIA

# Kiitos. Thank you.

Live: tasehinta.lovable.app

Frank Ponce · Abderahim Mojahid · Md Sakibul Alam Patwary

Pavel Samsonov · Pavlo Polish